Motor-Mediated Creativity: Bridging Embodied Skill Training and Digital Expression

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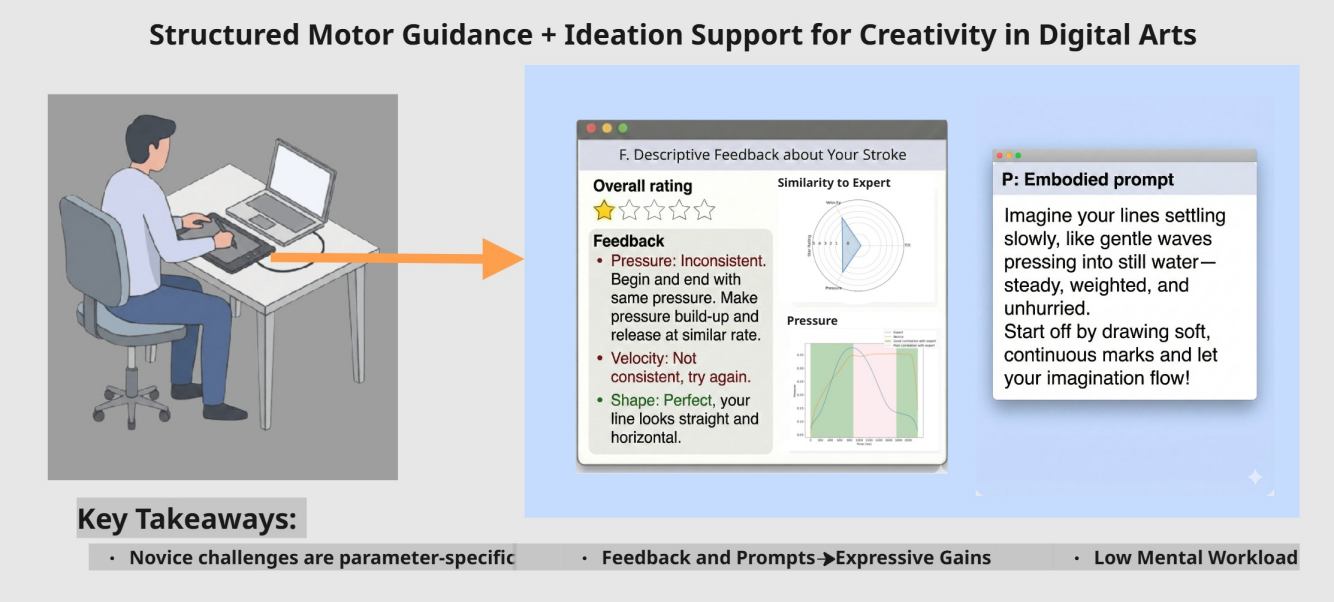


Figure 1: Teaser illustration of Motus, which combines structured motor guidance with embodied prompts to support expressive digital drawing. Novices receive multimodal stroke feedback (ratings, corrective text, radar, and temporal plots) alongside embodied prompts that frame drawing through sensory metaphors, helping bridge motor fluency and creative intent.

Abstract

Expressive digital drawing requires nuanced motor control, subtle variations in pressure, velocity, and rhythm that convey affect and style. While experts develop this embodied fluency through years of practice, novices struggle to produce marks that match their intentions, creating a gap between vision and execution. We propose motor-mediated creativity: treating motor training as integral to digital expression. Our system, Motus, instantiates this through structured practice of expressive primitives, expert-referenced feedback, and ideation prompts that encourage exploration. We report a two-stage investigation. A formative study characterized: (a) novice challenges in motor fluency, (b) examined how different feedback types, including corrective feedback, helped participants understand their mistakes, (c) how prompts, generic or embodied, support engagement with abstract expressive content. A controlled evaluation then linked fluency gains to subjective and expert ratings of

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expressiveness. Together, our findings show that scaffolding motor skills is a viable strategy for enhancing expressive agency in digital drawing.

CCS Concepts

• **Applied computing → Interactive learning environments**; • **Human-centered computing → Empirical studies in interaction design**; **Interactive systems and tools**.

Keywords

Motor-mediated creativity; digital art; corrective feedback; ideation guide; motor fluency; structured guidance

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1 Introduction

Digital creativity tools have made significant advances in recent years [13], integrating features such as AI-assisted drawing (e.g., automatic line completion, style transfer [20, 45]), procedural generation (e.g., automated synthesis of textures [16] or geometric forms), and physics-based simulation (e.g., brush deformation, ink diffusion [41]). Despite these computational improvements, these systems remain fundamentally constrained by a persistent bottleneck: the translation of creative intent into expressive output through motor input [13, 44].

Stylus-, pen-, and touch-based interfaces all require users to control stroke dynamics with nuance, subtle variations in velocity, pressure, directionality, and rhythm that are crucial for expressive quality [1]. For skilled experts, such embodied fluency can transform a simple line into an evocative mark. For novices and intermediates, however, the lack of motor precision [21] often produces outputs that feel flat, unintended, or frustratingly inconsistent. This gap between envisioned expression and executed mark reflects a sensorimotor bottleneck: while the digital system affords rich possibilities, users cannot always realize them due to limited motor fluency [2, 5].

Research in cognitive science and embodied cognition emphasizes that this bottleneck is not merely technical but deeply cognitive [40]. Motor actions encode affect, rhythm, and stylistic nuance; they shape conceptual reasoning and influence both aesthetic judgment and idea generation. Studies of gesture and movement highlight how physical action scaffolds ideation and emotional expression [24]. Motor learning research further demonstrates that explicit awareness of movement parameters, when supported with structured feedback, accelerates skill acquisition and facilitates transfer across tasks [23, 25]. In short, motor execution is not peripheral to creativity, it is integral to it.

Yet HCI research has rarely addressed this dimension. Creativity support systems in our field have largely emphasized expanding functionality, more brushes, more filters, more automation, while neglecting the embodied substrates of creative expression [10]. As a result, many users, particularly those transitioning from traditional media or those with strong visual sensibilities but weaker motor control, remain unable to fully leverage digital tools [28]. The absence of tactile resistance and material feedback in digital environments further exacerbates this disconnect, making it harder for users to internalize how motor actions map onto expressive outcomes.

We argue that addressing the sensorimotor bottleneck is a core interaction design challenge for creativity support. If creativity tools are to empower expressive practice, they must move beyond automating outcomes and toward scaffolding the embodied learning processes that underlie creative fluency.

Building on these observations, we designed Motus to directly address the two gaps surfaced in our formative work: (i) novices' difficulty in developing motor fluency, and (ii) their limited ability to translate intentions into expressive variation compared to experts. More specifically, we introduce the concept of motor-mediated creativity: a paradigm that enhances expressive digital output by explicitly cultivating motor fluency as a creative resource. Rather than bypassing motor effort through automation, motor-mediated creativity foregrounds intentional movement practice as a pathway to expressive agency. Our interactive system, Motus (a sample teaser is illustrated in Fig. 1), operationalizes this paradigm through two strategies:

- Explicit Motor Training: Structured practice of expressive primitives (e.g., confident strokes, pressure-modulated curves, rhythmic mark-making) paired with feedback calibrated to expert reference profiles.
- Ideation Prompts: Creative guides that encourage users to apply learned motor strategies in expressive, open-ended drawing tasks through sensorimotor metaphors and situational framings.

Our approach is grounded in embodied cognition, motor learning, and creativity studies. By explicitly surfacing the links between motor execution and expressive outcome, we aim to accelerate a process that expert artists acquire implicitly over years of practice: the internalization of motor–expression mappings (i.e., how movement dynamics such as speed or pressure shape the emotional quality of a stroke).

This paper makes three contributions to the field of HCI and creativity support:

- **A Design Paradigm for Motor-Mediated Creativity:** We introduce and instantiate motor-mediated creativity, a framework that integrates motor training, reflective transfer, and creative application as a pathway for enhancing expressive digital practice. This paradigm reframes creativity support from feature expansion to embodied skill cultivation.
- **Empirical Insights from a Formative Study:** Through a formative study with novice digital artists, we characterize the motor challenges users face when attempting expressive drawing and identify which forms of feedback and prompting are most effective for building motor–expression awareness. Together with reference profiles derived from expert artists, this analysis surfaces the motor parameters, such as velocity, pressure, and tilt, that most strongly influence perceived expressiveness.
- **System Implementation and Controlled Evaluation:** We present Motus, an interactive environment that operationalizes

motor-mediated creativity through multimodal feedback and embodied prompts. In a controlled evaluation study, we show that these interventions significantly improve novices' expressive fluency, creative agency, and the perceived quality of their digital artworks. This provides empirical evidence that explicit motor training and ideation support can jointly enhance creative outcomes in digital drawing, extending prior work on motor guidance.

This work positions motor-mediated creativity as a new trajectory for creativity support research. By centering embodied skill acquisition, it broadens the scope of how interaction design can foster expressive agency. Our findings suggest that digital creativity systems should not only expand what can be made, but also expand the human capacity to make expressively. We see this as a crucial step toward bridging embodied cognition, motor learning, and digital creativity in HCI.

2 Related Work

Our work builds on three main areas of prior research: (i) motor skill feedback in performance domains, (ii) feedback and guidance in digital drawing systems, and (iii) prompt-based ideation for creativity support. We position our contribution as extending these directions toward *motor-mediated creativity*, where motor fluency and ideation are jointly supported.

2.1 Creativity Support in HCI

Creativity support systems have been a longstanding area of interest in HCI, with early frameworks emphasizing the design of environments that enhance discovery, exploration, and reflection rather than automating output [29, 34]. In digital art and design, this trajectory has led to tools that expand the range of representational and stylistic possibilities through new features such as multiple brush engines, layer management, and more recently, AI integration.

AI-assisted creativity tools have advanced rapidly, enabling capabilities such as automatic line completion, neural style transfer, and text-to-image generation [11, 14]. Similarly, procedural generation techniques have been incorporated into art software to synthesize textures, forms, and complex patterns with minimal user effort [20]. These tools demonstrate the power of computation to extend what can be created, but they often prioritize efficiency and automation over embodied engagement.

A growing critique within HCI emphasizes that while such systems broaden access and expand functionality, they risk marginalizing the role of the human body and skill development in creative practice [15]. For example, while AI-based tools can generate high-quality outputs, users may feel less agency or expressive authorship when the system takes over the creative act [37]. This tension between automation and agency is particularly salient in art-making, where fluency in physical mark-making is central to expressive identity.

Our work builds on this line of critique by focusing on the *sensorimotor bottleneck* – the difficulty many users face in translating creative intent into expressive digital marks. While HCI research has explored interaction techniques for digital drawing (e.g., calligraphy training systems, stylus-gesture interaction), relatively little attention has been given to cultivating motor fluency as a first-class design goal in creativity support. By situating motor training and embodied ideation as integral to creative systems, our approach seeks to complement existing functionality and AI-focused directions with support for the embodied processes that underlie expressive practice.

2.2 Embodied Cognition and Motor Learning

Cognitive science and psychology provide strong evidence that bodily movement is integral to thinking and expression. Theories of embodied cognition argue that motor actions are not merely instrumental but encode affective and conceptual information that shapes perception, memory, and creativity [4, 42]. Motor learning research demonstrates that explicit attention to parameters such as velocity, force, and tilt can accelerate skill acquisition and transfer across tasks [19, 26]. Neurocognitive studies show that motor imagery activates neural pathways similar to physical execution, suggesting that conscious reflection on movement can enhance both execution and ideation.

Within art and design, expertise studies reveal that skilled artists exhibit distinctive motor signatures that underpin expressive effects. Yet such motor fluency is typically acquired only through extended implicit practice. This body of work motivates our focus: making motor–expression mappings explicit so that users can intentionally deploy motor control as a creative resource.

2.3 Feedback and Scaffolding in Digital Learning Systems

Feedback and scaffolding have long been recognized as central mechanisms in technology-enhanced learning. Foundational work in educational psychology highlights that feedback functions not only as an error-correction mechanism but also as a driver of metacognitive awareness and self-regulation [35]. In digital learning environments, scaffolding typically refers to the provision of adaptive supports, such as hints, visualizations, or worked examples, that are gradually withdrawn as learners gain proficiency [43].

Feedback-driven training has been extensively explored in related motor domains such as handwriting [38], music performance [22], massage instruction [17], AR-based instructional system [46], and sports training [36]. Across these areas, researchers have shown that different forms of feedback, whether through visual overlays, multimodal cues, or structured performance metrics, can significantly support the development of fine-grained motor skills.

Within creativity support systems, however, feedback and scaffolding have largely emphasized conceptual or procedural guidance (e.g., idea generation prompts, tutorial-style interfaces) rather than the cultivation of embodied motor fluency. Prior work on digital art pedagogy, for instance, has introduced systems for guided practice or step-by-step tutorials, but rarely addresses the fine-grained feedback needed to help novices link motor execution to expressive outcomes. Our work extends this line by designing feedback and scaffolding that explicitly target sensorimotor processes—providing users with interpretable multimodal feedback on stroke dynamics, coupled with embodied prompts that scaffold ideation. In doing so, we reposition feedback not just as an instructional tool, but as a

bridge between physical action and expressive agency in digital drawing.

2.4 Prompt-Based Ideation for Creativity Support

Prompting has become a common strategy for supporting ideation in digital creativity tools, from thematic prompts to AI-generated suggestions. However, most prompts focus on conceptual or stylistic inspiration without addressing the embodied dimension of expression.

Our system contributes by introducing *embodied ideation prompts*. These prompts are metaphorical or sensorimotor in nature (e.g., "draw how it feels to stretch after waking up," "make marks like footsteps on different surfaces"), encouraging users to connect motor action with expressive intent. By comparing embodied prompts against generic technique-oriented prompts, we directly test whether embodied framings more effectively activate motor strategies during creative work.

2.5 Theoretical Integration: Motor-Mediated Creativity

Motor-mediated creativity synthesizes embodied interaction [6], motor learning theory [33], and creativity support [34]. We position our work by contrasting it with two prevailing approaches in embodied CSTs. Systems like *Sketch-Sketch Revolution* [8] gamify *reproductive accuracy* (tracing), prioritizing fidelity to a target over personal expression.

We address this gap by operationalizing motor learning principles, specifically Schmidt's schema theory [32], to support *expressive fluency*. Unlike automation or tracing tools, Motus treats fine-grained motor control (pressure, tilt, velocity) as a trainable expressive resource. By making these parameters explicit through feedback and linking them to ideation via embodied prompts, we bridge the gap between technical execution and creative intent.

Our findings provide systematic empirical evidence that explicit motor training can enhance creative outcomes in digital drawing. This extends the scope of creativity support beyond feature expansion to include the cultivation of embodied skill.

2.6 Summary: Gap and Contribution

Across embodied cognition, motor skill training, and creativity support, prior work has established that movement underlies both execution fluency and ideation. Yet existing systems tend to address either motor accuracy (e.g., sports, healthcare) or idea generation (e.g., prompt-based creativity tools), rarely bridging the two. What remains underexplored is *motor-mediated creativity*: cultivating embodied control of stroke dynamics as an expressive resource, and scaffolding its deployment in open-ended creation. Our system directly addresses this gap by combining interpretable motor feedback with embodied ideation prompts and by evaluating their joint impact on both motor fluency and creative expressiveness.

3 Stroke Taxonomy for Expressive Motor Training

To support motor fluency and expressive awareness in digital drawing, we curated a set of twelve canonical strokes (detailed in Table 7) derived from traditional drawing pedagogy, expert teaching resources, and analysis of expressive mark-making techniques [7, 27]. These strokes are not representational but function as *expressive primitives*: simple gestures that isolate key motor parameters while fostering transferable skills for creative expression.

Our taxonomy is grounded in theories of motor learning and drawing pedagogy, which emphasize progression from isolated parameter control to integrated, multi-modal execution [7, 27]. These strokes function as expressive primitives, structured into five categories ranging from basic pressure dynamics to complex multi-modal integration. A detailed pedagogical rationale is provided in Supplementary Materials.

Pedagogical Validation and HCI Relevance. This taxonomy was refined in collaboration with ten professional digital illustrators (at least with 2-3 years of professional experience or a formal diploma/degree in Fine Arts), and one art pedagogy instructor, who confirmed that the primitives align with both art pedagogy and natural motor development. Our formative study further validated its relevance: novices struggled most with pressure control and directional transitions, precisely the areas emphasized in the early categories.

4 Expert Data Collection

Although the 12 canonical strokes were predefined based on drawing pedagogy, we conducted an expert study to (i) construct high-quality execution profiles for use in our feedback system and (ii) gather pedagogical insights into which strokes are most critical for developing expressive fluency. These data informed both the design of Motus and our interpretation of novice performance in the formative study. This expert data collection is approved by our institute's IRB.

4.1 Method

We recruited ten professional illustrators (6 female, Age: M= 24.36, SD = 8.71) (at least 2-3 years of professional experience as a digital artist or a formal education in Fine Arts) and asked them to execute the full set of 12 expressive strokes using a Wacom Movink 13 display and pressure-sensitive stylus [1]. Wacom Movink 13 has a pressure sensitivity of 8,192 levels and the stylus motion was sampled at 200Hz capturing x-, y-position, tilt, rotation, and pressure. For each stroke, experts completed three repetitions to provide stable motor traces.

Following stroke execution, experts completed structured questionnaires and participated in semi-structured interviews. We probed their perspectives on the pedagogical role of each stroke, including which ones they considered foundational for novices, which were more suitable for advanced training, and how they themselves use such strokes to communicate expressive nuance in professional practice.

[1] https://www.wacom.com/en-us/products/pen-displays/wacom-movink

Stroke	Visual	Description	Targeted Motor Skill
Pressure Crescendo		4-inch line; pressure rises to a midpoint peak, then releases.	Creates emphasis and dramatic variation in line weight.
Pressure Staccato		8 short marks alternating light and heavy pressure.	Rapid switching between force levels with spatial precision.
Pressure Sustain		3-inch line with constant maximum pressure.	Sustained heavy-force control and directional stability.
Accelerando Line		4-inch line with steadily increasing speed.	Coordinated acceleration while maintaining path accuracy.
Rhythmic Pulses		6 connected curved pulses drawn in steady rhythm.	Timed repetition with smooth directional shifts.
Confident Curve		Single smooth S-curve, no corrections.	Continuous directional transitions with motor fluency.
Angular Precision		Zigzag of 6 segments at exact 60° angles.	Sharp, controlled directional changes.
Parallel Consistency		5 horizontal lines with perfect 0.25in spacing.	Consistent spatial alignment across repetitions.
Responsive Curves		Paired curves in spatial "dialogue" — echoing or contrasting.	Sequential planning and relational placement.
Directional Pressure Shift		Diagonal stroke fading from heavy to light pressure.	Simultaneous control of angle and graded pressure.
Temporal Spatial Spiral		Outward spiral: fast and tight at the center, slowing as it expands.	Coordinated deceleration with radius growth.
Rhythmic Wave		4 undulating waves with consistent weight.	Consistent pressure with curvature timing.

Table 1: Stroke taxonomy showing images of strokes, execution details, and targeted motor skills.

Due to a technical malfunction in the data acquisition system, the stroke data from one expert were incomplete. Consequently, the analysis was conducted with a sample of nine experts.

4.2 Findings

Experts generally affirmed the pedagogical relevance of all 12 strokes but offered three key insights:

- **Pressure modulation is central to expressivity.** All the experts identified pressure variation as the most influential factor in shaping line quality and affective tone. *E9* explained, "strokes that reflect pen pressure or have either sharp or curved edges help me depict different moods and emotion in the line work process." *E5* added that while maintaining constant pressure felt "unnatural" for their own practice, they could see its value: "the only thing new and unnatural is the same pressure point throughout. So I might use that technique to do simple line art as a creative art style, maybe Chibi styles."
- **Temporal rhythm conveys vitality.** Experts described acceleration, deceleration, and rhythmic repetition as central to conveying energy and emotion. *E4* noted, "curves are the most important strokes to me, as my art style is more sketchy and dynamic – I like fluidity in art, hence curves help to express that the most. Yes, I would pay more importance to stroke speed and velocity." *E5* reflected on their own experience, saying, "when I'm agitated I draw spiky things like stroke 7. Spiral helps me to express when I'm stuck or bored in class or just want to express my curiosity for the unknown."
- **Integrated control emerges gradually.** Experts emphasized that more complex strokes, those requiring simultaneous modulation of pressure, speed, and directionality, should be introduced only after novices gain fluency with simpler exercises. *E9* explained, "I think the main importance of these strokes is to build up confidence in the learner rather than to get them perfect looking." *E5* highlighted the importance of confidence building through rhythmic variation, describing how mural projects and art therapy contexts often use themed prompts: "I used the Emotion Wheel before—it gives a theme or premise for people to actually draw out emotions."

These insights reinforced the staged progression embedded in our taxonomy: starting with pressure control, layering in rhythm

and directionality, and eventually moving toward integrated multi-parameter fluency. They also provided validation that expressive strokes are not only technical drills but pedagogical tools for creating confidence and expressive awareness.

5 Formative Study

To inform the design of our motor-mediated creativity system, we conducted a formative study with novice digital artists. The study was designed to uncover which motor features most affect expressive outcomes, and to examine two key forms of support. By feedback, we refer to system-generated responses that help users understand and improve their motor execution (e.g., visual comparisons with expert profiles or descriptive textual guidance). By prompts, we refer to task framings that spark and guide creative exploration (e.g., embodied metaphors or scenario-based instructions that connect movement to expressive intent). Together, these elements allowed us to investigate what forms of system support are most interpretable for novices and how different framings influence their ability to translate intention into expressive digital marks. Both the formative and the evaluation study (detailed in Section 7) are approved by our institute's IRB.

5.1 Research Questions

We investigated two questions:

- **RQ1 (Novice Challenges):** What are the primary difficulties novices face when attempting expressive drawing in digital environments? Which motor parameters (e.g., velocity, pressure, timing, directionality) are most difficult for novices to control during expressive drawing?
- **RQ2 (Feedback and Prompt Effectiveness):** How do different feedback modalities and prompt framings influence novices' ability to improve motor control and develop expressive stroke strategies across repeated attempts?

5.2 Participants and Apparatus

We recruited 42 participants (25 male; Age: $M = 26.65$, $SD = 3.72$), all beginners in digital drawing with no formal visual arts training. Participants reported an interest in creative expression but self-identified as novices in both digital and traditional media.

Sessions were conducted on the same device we used to collect the expert data – a Wacom Movink 13 display with a pressure-sensitive stylus (8,192 pressure levels and 200Hz sampling rate). A custom-built web-application employing the data processing explained in Section 5.4 is used. The interface displayed strokes in standard drawing view and supported feedback visualizations.

5.3 Procedure

Each session lasted approximately 90 minutes and included four components:

Pre-Study Questionnaire. Participants first completed a structured questionnaire probing prior drawing experience, difficulties, and expectations. Items included 5-point Likert-scale ratings of difficulty with pressure, velocity, and directionality, alongside open-ended questions on frustrations with digital tools.

Phase 1: Motor Training with Feedback Conditions. Participants began with stroke-based motor training exercises focusing on three features identified as central to expressive line quality: velocity ramps, pressure modulation, and directional transitions. Prior work highlights these parameters as particularly influential in how viewers perceive expressiveness.

Each participant practiced under four different feedback conditions, presented in counterbalanced order:

- **Reference Overlay.** A visual overlay compared the participant's stroke to an expert profile. This design draws from motor training in domains like sports and music, where comparison visualizations help learners quickly spot deviations.
- **Descriptive Comparison.** In this condition, the system generated short textual summaries that (a) highlighted how a participant's stroke diverged from an expert reference and (b) evaluated each stroke against its intended feature profile. To derive these summaries, we implemented a rule-based scoring pipeline: (i) for strokes with a target orientation, we measured the angular deviation of the trajectory to assess whether the line remained horizontal or vertical; (ii) smoothness was quantified by analyzing local velocity and curvature, with discontinuities flagged as roughness; (iii) pressure and velocity profiles were compared against the expected trend (e.g., increasing, decreasing, or sustained) defined for each stroke type. Each feature was then color-coded as positive (green) or negative (red), and the most salient deviations were converted into natural-language feedback (e.g., "You released pressure too early, resulting in a flatter arc" or "Your line is slanting upward instead of staying horizontal"). We included this condition because verbalized guidance, grounded in interpretable stroke features, offers novices direct and actionable suggestions while reducing the burden of interpreting abstract numerical scores.
- **Generic Metrics.** Numerical data such as average velocity or pressure variance were displayed without interpretation. This baseline helped us examine whether raw statistics alone can meaningfully support motor learning in creative contexts.
- **No Feedback.** Participants simply saw their drawn strokes with no additional information, simulating unguided practice in most drawing tools.

Phase 2: Expressive Drawing with Prompt Conditions. In the second phase, participants engaged in open-ended drawing tasks designed to surface how different kinds of prompts shaped their expressive output. Participants were asked to complete a set of abstract mark-making tasks designed to elicit expressive variation across different affective states. We selected Joy, Calm, Tension, and Chaos to span the arousal-valence spectrum of Russell's Circumplex Model [30]. We prioritized these states because they elicit distinct, measurable kinematic signatures (e.g., velocity and pressure changes) necessary for evaluating motor feedback. Low-arousal/negative states (e.g., sadness) were excluded due to their association with motor inhibition, which offers a limited dynamic signal for assessing fluency improvements. Rather than drawing objects or figures, participants were explicitly instructed to convey these states through non-representational means such as lines, textures, and rhythmic marks. This framing emphasized the motor–expressive dimension of drawing, allowing us to examine how novices used stroke dynamics alone to translate affective intent into visual form. We compared three prompt conditions:

Emotion	Formative Study (Generic Artistic Prompt)	Formative Study (Embodied Prompt)	Evaluation Study (Embodied Prompt)
Happiness	Create dynamic marks with varied line weights and energetic movement.	Express bursting happiness! Imagine your marks are stretching outward, expanding like laughter filling a room.	Imagine your lines bursting outward like sparks, stretching, swirling, and overlapping with vibrant energy. You may start off by drawing dynamic marks and let your imagination speak for the rest!
Calm	Make smooth, continuous marks with consistent speed and steady pressure.	Express deep serenity. Imagine dragging a heavy brush slowly through soft sand.	Imagine your lines settling slowly, like gentle waves pressing into still water—steady, weighted, and unhurried. Start off by drawing soft, continuous marks and let your imagination flow!
Tension	Make irregular, broken marks with sudden stops and changes in direction.	Express nervous energy. Imagine your marks pacing back and forth, unable to settle, like tapping your foot restlessly.	Imagine your lines twitching in place, stopping and starting, pressing against each other without release—uneven, sharp, and unsettled. You may explore marks that are jittery and let your imagination flow!
Chaos	Make overlapping, dense marks with rapid changes in direction and pressure.	Express total overwhelm. Imagine your marks colliding and interrupting each other, like tangled voices in a crowded festival.	Imagine your lines rushing in from every direction, crossing, piling up, and tangling like a storm with no center. Let them spill and clash until the page feels overrun. You can start off by exploring dense mark making and let your imagination speak for the rest!

Table 2: Examples of prompts used in formative and evaluation studies across four target emotions.

- **No Prompt**. Participants were simply told to "make expressive marks" for a given emotion. This served as a baseline for spontaneous exploration, capturing how novices approach expression without external guidance.
- **Generic Artistic Prompt.** Here, participants received neutral, technique-oriented instructions such as "draw lines with different thicknesses" or "create smooth and jagged marks." These mirror exercises common in art education, intended to encourage variation but without explicitly invoking affect or imagination (tabulated in Table 2).
- **Embodied Prompt.** Participants received metaphorical instructions tied to bodily or sensory experiences – for example, "Express bursting happiness! Imagine your marks are stretching outward, expanding like laughter filling a room" or "Express total overwhelm. Imagine your marks colliding and interrupting each other, like tangled voices in a crowded festival." We included this condition because prior work suggests that embodied metaphors can spark ideation and help novices connect motor actions with expressive outcomes (tabulated in Table 2).

Post-Study Questionnaire and Interview. Participants rated the helpfulness of each condition on 7-point Likert scales and ranked conditions by preference. Semi-structured interviews probed perceived learning, strategies, and awareness of motor–expression mappings.

5.4 Data processing

In custom web-based drawing system, stylus pressure obtained via JavaScript is delivered through the Pointer Events API, which provides a device-independent and normalized measurement. Irrespective of the stylus hardware or sensor resolution, the API scales the underlying raw pressure values to a standardized interval between 0 and 1, where 0 represents the absence of contact and 1 denotes the maximum pressure the device can register. Although this normalization promotes cross-device comparability, it does not fully eliminate variations in how different devices convert physical force into the normalized scale. Following data capture, the pressure values together with the timestamp, x- and y-coordinates, and the tilt components in the x- and y-directions are exported to a JSON file. As an initial processing step, data points of last *pointerup* events are removed, as they correspond to the interval between the completion of the stroke and tapping on the *save* button. For subsequent analyses, absolute values are computed for both velocity and tilt. The standard deviations of pressure, velocity, and tilt are then used as proxies for motor inconsistency. Under this assumption, larger standard deviation values indicate greater variability in the respective parameter, and thus reflect higher levels of inconsistency.

When a direct stroke-level comparison is required between individual users or groups of users, each time series is first one-dimensionally interpolated to a common temporal scale. After temporal alignment, the Pearson correlation coefficient is computed to quantify the similarity between the strokes.

5.5 Findings

Although preliminary, the study revealed consistent patterns across conditions that shaped our system design.

5.5.1 RQ1: Novice Challenges in Expressive Digital Drawing. To investigate the primary challenges novices face when attempting expressive drawing in digital environments, we analyzed both quantitative measures of motor fluency and qualitative reports from post-study questionnaires.

Our analysis revealed three core challenges novices face when engaging in expressive mark-making with a stylus interface: (i) motor control instability, (ii) difficulty linking motor actions to intended expressive qualities, and (iii) managing competing demands between technique and intended expression.

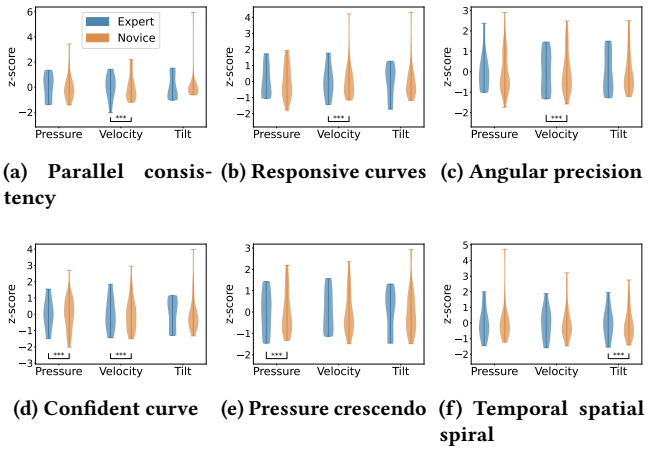
Challenge 1: Difficulties in Motor Control. To assess how novices differ from experts in controlling motor features, we analyzed the variance of inconsistency in stroke execution using Levene's test. Given that the data were non-normally distributed (per Shapiro–Wilk tests, $p < 0.05$), this approach allowed us to examine whether novices exhibited greater inconsistency in pressure, velocity, and tilt – an indicator of motor difficulty. Our premise was that if novices matched expert fluency, their inconsistency would be statistically indistinguishable from expert traces.

To visualize the stability of novice motor control relative to professionals, we converted raw inconsistency metrics into z-scores standardized against the expert baseline. In Fig. 2, the y-axis represents motor instability; values closer to zero indicate consistent, expert-like execution, while higher values indicate erratic control. The visual contrast between the groups is distinct: Expert profiles (blue) are tightly clustered, reflecting high reproducibility. In contrast, Novice profiles (orange) exhibit elongated, vertically distributed shapes, statistically confirming that novices struggle to maintain consistent pressure and velocity (higher z-scores), particularly during spatially constrained tasks.

- Velocity & Spatial Precision: Velocity control was the most persistent difficulty, with significantly higher novice variance across 7 of 12 strokes, particularly in spatially constrained tasks like Parallel Consistency ($W(1, 49) = 6.33, p < 0.05$) (Fig. 2a) and Responsive Curves ($W(1, 49) = 4.19, p < 0.05$) (Fig. 2b). In addition to that in Angular Precision ($W(1, 49) = 4.75, p < 0.05$) (Fig. 2c) and in Confident Curve ($W(1, 49) = 5.29, p < 0.05$) (Fig. 2d) where directionality is crucial.
- Pressure Control: Novices exhibited significantly higher variance in strokes requiring gradual modulation, such as Confident Curve ($W(1, 49) = 4.55, , p < 0.05$) (Fig. 2d) and Pressure Crescendo ($W(1, 49) = 4.49, p < 0.05$) (Fig. 2e).
- Multi-Parametric Integration: Complex strokes requiring simultaneous tilt and pressure modulation (e.g., Temporal Spatial Spiral depicted in Fig. 2f) showed significant deviation in tilt ($W(1, 49) = 5.14, p < 0.05$).

Qualitative Results: Qualitative feedback corroborated these quantitative gaps. Participants identified three core hurdles: (i) Motor Instability, where users struggled to "control muscle to draw smoothly" (P6) or manage "pressure and velocity at the same time" (P2); (ii) Intention-Action Gaps, describing a disconnect where they knew what they wanted but "were not able to make [their] hand do it" (P20); and (iii) Cognitive Load, noting that balancing technical precision with expression felt like "divided attention" (P26). For more details on these challenges, refer to the Supplementary Material.

RQ1 Summary. These findings indicate that novices' difficulties are multi-layered: unstable motor control, weak motor-to-intent mappings, and the challenge of balancing technical execution with intended expressivity. These insights highlight the need for feedback systems that scaffold motor precision while simultaneously supporting learners in developing an embodied understanding of expressive stroke parameters.



(a) Parallel consistency **(b) Responsive curves** **(c) Angular precision**

(d) Confident curve **(e) Pressure crescendo** **(f) Temporal spatial spiral**

Figure 2: Stroke-level motor performance across parameters. Higher values indicate greater difficulty (e.g., higher variance, lower consistency). The legend provided in panel (a) applies to all sub-figures.

5.5.2 RQ2: Feedback and Prompt Effectiveness. Post-study rankings revealed a clear hierarchy in preferred support (Table 3).

As shown in Table 3, participants overwhelmingly favored Descriptive Textual Feedback (37/42) and Embodied Prompts (33/42), citing their interpretability and ability to bridge the gap between intent and execution.

Participants recommended two major improvements, which were incorporated into the final design. First, they suggested combining descriptive text with visual overlays to better "elaborate on mistakes" (P5) and pinpoint specific error zones (P10). Second, they requested quantitative performance metrics, such as an overall "similarity score" (P12) or specific parameter ratings (P17), to benchmark their drawing quality.

5.6 Design Considerations

From the formative study, we derive several design considerations that directly shaped the development of our motor-mediated creativity system.

First, interpretable feedback emerged as essential for novice learning. While raw numerical metrics were rarely actionable, participants found visual overlays and textual descriptions effective for understanding how their motor actions diverged from expressive standards. This suggests that feedback mechanisms should prioritize clarity and direct comparability over quantitative precision. Building on participant suggestions, we adopted a multimodal feedback design that combines complementary channels: concise textual guidance to highlight errors and provide actionable corrections, visual overlays to illustrate deviations, and an overall similarity score to give learners an accessible performance summary. Together, these layers balance interpretability, diagnostic

Condition Category	Intervention Type	Top Choice (Count)	Qualitative Rationale (Summary)
Motor Feedback	Descriptive Text	37	"Intuitive," "actionable," "like a human tutor"
	Reference Overlay	5	"Useful for diagnosis," but "high cognitive load"
	Generic Metrics	0	"Opaque," "useless," "hard to interpret"
Ideation Prompts	Embodied Prompts	33	"Sparked imagination," "bridged feeling to stroke"
	Generic Prompts	9	"Clear instruction," but "limited exploration"

Table 3: Participant preferences across intervention types, with top-choice counts and qualitative rationales.

depth, and motivational clarity, ensuring that novices can both understand their performance and identify concrete next steps for improvement.

Second, embodied prompts proved especially valuable in stimulating intentional variation. By framing drawing tasks through bodily sensations or metaphorical actions, participants were able to more readily connect motor execution with expressive output. In contrast, generic technical prompts or unguided practice offered less engagement and less noticeable variation. This indicates that ideation supports in creative systems should leverage embodied metaphors rather than focusing solely on technical variation.

Third, the findings highlight that pressure modulation and velocity smoothness are the most impactful motor parameters for expressive quality. Systems aiming to scaffold expressive drawing should therefore emphasize these features in both training exercises and feedback, rather than attempting to cover all possible dimensions of stroke dynamics equally.

Finally, participants frequently expressed frustration with the gap between intention and outcome. This reinforces the importance of designing interfaces that make motor–expression mappings more transparent, enabling learners to form stronger intuitions about how their actions translate into visual results.

6 Motus: System Design

Building on insights from the formative study, we designed Motus, an interactive drawing environment that scaffolds motor-mediated creativity through two complementary mechanisms: (i) supporting motor fluency with interpretable, multimodal feedback, and (ii) stimulating creative ideation through embodied prompts. The system integrates structured motor training with open-ended creative tasks, providing both parameter-level feedback and inspiration for expressive exploration. The overall architecture of our Motus is illustrated in Fig. 3.

6.1 Design Goals

Findings from the formative study revealed two persistent challenges for novice digital artists: (i) difficulty controlling fine-grained motor parameters such as pressure and velocity, and (ii) difficulty linking creative intentions to expressive outcomes. Participants reported frustration at "knowing what I want to draw but not being able to make my hand do it." These insights informed two design goals for Motus:

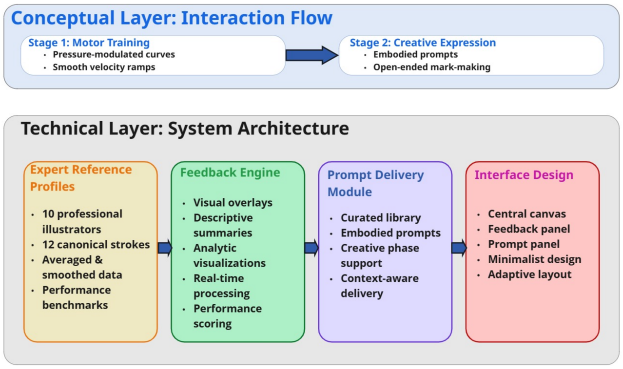


Figure 3: Motus: System Overview

(1) **Motor fluency support:** Help users build awareness of how motor actions shape expressive outcomes by surfacing feedback that is both interpretable and actionable.
(2) **Embodied ideation scaffolds:** Encourage creative exploration through embodied prompts that link drawing tasks to sensory or metaphorical experiences.

6.2 Conceptual Layer: Interaction Flow

Motus follows a two-stage interaction flow. In the first stage, users practice predefined expressive stroke primitives identified in the formative study, such as pressure-modulated curves, smooth velocity ramps, and directional transitions. These primitives were consistently highlighted by experts as central to expressive fluency, yet difficult for novices to master. In the second stage, users transition into creative mark-making tasks framed by embodied prompts. This progression allows learners to refine parameter-level motor control before applying it in open-ended expression.

6.3 Technical Layer: System Architecture

Motus was implemented as a standalone drawing application on a Wacom Movink 13 tablet with a pressure-sensitive stylus. Data processing is performed as explained in the Section 5.4 with 200Hz sampling rate and 8,192 levels pressure sensitivity.

Expert Reference Profiles. We collected stroke data from nine professional illustrators performing the 12 canonical strokes described in Section 3. These data were averaged and smoothed to generate expert reference profiles, which serve as comparison anchors for feedback generation.

Feedback Engine. The feedback engine integrates expert reference profiles with user traces to generate multimodal feedback. Feedback is delivered through a combination of visual overlays, descriptive summaries, and analytic visualizations. Details of how this feedback was operationalized in the user study are provided in Section 7.

Prompt Delivery Module. Embodied prompts are organized into a curated library designed by the research team. These prompts act as affective triggers, analogous to the use of music in traditional art pedagogy to induce specific moods. Their role is to stimulate the motor system for expressive execution, rather than serving as the primary object of study. Prompts are presented during the creative phase, where users apply learned motor strategies in open-ended tasks.

Interface Design. The interface is structured around three regions: a central canvas for drawing, a feedback panel activated during motor training, and a prompt panel active during creative exploration. The design emphasizes minimalism, ensuring that feedback and prompts are accessible without distracting from the drawing process.

7 User Evaluation

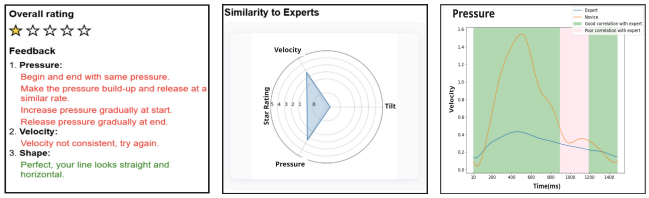
Building on insights from the formative study, we conducted a controlled evaluation to examine how motor feedback and ideation prompts contribute to expressive skill development in digital drawing. The evaluation focused on disentangling the effects of each intervention component and testing the combined effectiveness of Motus as a motor-mediated creativity support system.

7.1 Participants

We recruited 36 participants (17 female; age range 17–32, $M = 23.69$, $SD = 4.19$), restricting to novices with no formal training in digital drawing and minimal prior stylus experience. For this phase, we randomly assigned 36 participants to one of four experimental groups (G1–G4) of equal size.

Rationale for Participant Cohort and Design: Participants for the controlled evaluation were drawn from the pool who completed the formative study one week prior. This design choice was deliberate: the formative phase allowed us to acquire crucial, individualized baseline data on each novice's motor profile (RQ1 challenges), ensuring that the subsequent factorial evaluation effectively measured differential learning gains (Δ Post-Pre) over a constant baseline exposure, thereby maximizing statistical power for the comparative intervention effects. Further, the formative phase involved only exploratory exposure to components of the feedback system (e.g., metrics alone, or descriptive text alone) and heterogeneous prompt framings, guaranteeing that participants had not yet experienced the final, integrated, multimodal intervention tested in the controlled evaluation. We have conducted and included an additional validation study (Section 9) with a new set of users to further verify our findings.

Hardware Generalization Check: To ensure the robustness of the core findings against external limitations (i.e., hardware specificity and carry-over effects outside the initial cohort), we conducted a focused validation study using an independent sample and an alternate platform (iPad Pro), which is reported separately in Section 9.



(a) Similarity score and corrective feedback **(b) Radar plot showing similarity to expert** **(c) Temporal variation plot**

Figure 4: Multimodal Feedback Design Informed by the Formative Study.

7.2 Study Design

The experiment employed a between-subjects 2 × 2 factorial design, crossing the presence or absence of (1) motor feedback and (2) ideation prompts. The resulting groups were:

- G1 (Control): no motor feedback, no ideation prompt
- G2 (Prompt Only): embodied ideation prompts without motor feedback
- G3 (Feedback Only): multi-modal motor feedback without prompts
- G4 (Full Motus): both motor feedback and ideation prompts

This design allowed us to test the independent contribution of each intervention and its combined effects.

7.3 Motor Feedback Intervention

Participants in G3 and G4 received a multi-modal motor feedback after each stroke attempt (illustrated in Fig. 4. Each component was chosen to make specific aspects of the motor–expression relationship explicit, targeting both immediate error correction and longer-term awareness of expressive parameters:

- Similarity Score (Star Rating): A one–to–five star rating quantified overall similarity to expert reference profiles. This simple summary metric was chosen to provide users with an accessible, at-a-glance indicator of performance, lowering cognitive load and motivating improvement.
- Corrective Feedback: (Textual Guidance): Contextualized text messages highlighted specific execution errors and offered actionable suggestions (e.g., "You lifted off too early – try maintaining steady pressure throughout"). This component addressed the need for direct interpretability, bridging raw metrics with concrete strategies that novices could immediately apply
- Radar Plot (Feature-Level Diagnosis): A multi-dimensional visualization displayed similarity scores across individual motor features (velocity, pressure control, smoothness, directional accuracy). This representation was included to help learners understand which aspects of their motor execution were strong or weak, supporting targeted practice and self-regulation.
- Temporal Variation Plot (Error Localization): For the weakest-performing feature, a temporal trace compared the novice's

stroke against the expert's, segmented into "good" and "poor" regions. This representation provided fine-grained temporal awareness, enabling participants to pinpoint when execution breakdowns occurred and linking moment-to-moment actions with expressive outcomes.

7.4 Ideation Prompt Intervention

Participants in G2 and G4 received embodied ideation prompts designed to encourage expressive exploration without constraining creativity. The prompt library was author-generated to function as a standardized affective prime. In traditional art pedagogy, educators often use music to immerse students in a specific emotional state (e.g., playing fast, chaotic jazz to encourage energetic mark-making). Our embodied prompts serve a similar function in the absence of audio: they use kinematic metaphors (e.g., "bursting," "flowing") to induce the intended arousal-valence state. Embodied prompts ground abstract creative intentions in physical or sensory experience (e.g., "Draw how it feels to stretch after waking up"), which novices reported made it easier to translate feelings into motor actions.

However, formative findings also revealed a tension: while some participants found generic prompts helpful because they offered clear, concrete instructions, others noted that such prompts felt prescriptive and reduced opportunities for exploration. In practice, generic prompts often led participants into controlled reproduction rather than open-ended expression.

To address this, our intervention combined embodied prompts with a minimal exploratory hint (e.g., "You may begin with smooth, continuous lines" or "Try fragmented strokes and let your creativity flow"). These hints acted as teasers, giving novices an accessible entry point into the task while avoiding over-specification. By suggesting a starting direction rather than dictating a procedure, the prompts were intended to spark creative engagement while preserving autonomy (the prompts used in the study are presented in Table 2, last column).

This design thus balanced two needs identified in the formative phase: (i) the expressive grounding afforded by embodied metaphors, and (ii) the practical guidance novices sought when unsure how to begin.

7.5 Tasks

Participants practiced the same set of 12 fundamental strokes introduced in the formative study, covering variations in velocity, pressure, direction, and rhythm. These strokes served as training primitives for developing motor–expressive fluency.

For the creative phase, participants completed four non-figurative expression tasks targeting distinct affective states: joy, calm, chaos, and tension (two per arousal–valence quadrant). As in the formative study, participants were instructed to convey these emotions through abstract marks only (e.g., lines, textures, rhythms) rather than representational imagery.

By pairing structured stroke training with open-ended creative challenges, the study design enabled us to examine both parameter-level motor improvements and their expressive transfer into holistic creative tasks.

7.6 Procedure

Each evaluation session lasted approximately 60 minutes and followed four stages:

- Baseline Creative Task (Pre-treatment Creativity): Participants first completed the set of four expressive drawing tasks (joy, calm, chaos, tension) without receiving any prompts or feedback. This served as a baseline measure of participants' "unguided expressiveness" and established their initial ability to convey affective states through abstract mark-making.
- Motor Training (Conditioned Practice): Participants then engaged in structured stroke training exercises, practicing the 12 fundamental strokes introduced in the formative study (each stroke was attempted five times). Depending on group assignment, participants received either no feedback, ideation prompts only, motor feedback only, or the full Motus intervention (both feedback and prompts).
- Post-treatment Creative Task: Participants repeated the four expressive drawing tasks. For groups assigned to receive ideation prompts, embodied prompts with exploratory hints were provided at this stage. This phase allowed us to assess how training and guidance influenced participants' expressive output relative to their baseline work.
- Survey and Interview: Finally, participants completed a post-study questionnaire assessing perceived expressiveness, learning, workload, and usability. A short semi-structured interview captured qualitative reflections on their experience, perceived challenges, and strategies for expression.

7.7 Measures

To evaluate the efficacy of Motus, we employed a mixed-methods framework by carefully choosing metrics that capture motor fluency using high-frequency log analysis and subjective expert assessments. All stylus input data (position, pressure, tilt) were sampled at 200Hz. Prior to metric computation, raw trajectories were smoothed using a 4th-order low-pass Butterworth filter (cutoff frequency = 6Hz) to attenuate physiological tremor and digitization noise while preserving voluntary movement dynamics. We categorized our evaluation measures into objective technical proficiency, subjective expressive quality, and user experience.

- **Technical Proficiency (Objective):**
  - Motor Fluency and Smoothness (The Quality of the Movement):
    - Spectral Arc Length (SPARC) [3]:To robustly quantify movement smoothness, we utilized the SPARC metric on the stroke velocity profile. Unlike dimensionless jerk, which is sensitive to measurement noise and duration, SPARC analyzes the Fourier magnitude spectrum of the speed signal. A lower spectral complexity (fewer high-frequency components) indicates smoother, more ballistic sub-movements characteristic of expert motor control.
    - 2/3 Power Law [18, 39]: We assessed biological motion fluency by calculating adherence to the 2/3 Power Law, which dictates that tangential velocity decreases as curvature increases. We extracted the log-log relationship between tangential velocity and curvature ($\log v(t) = \alpha - \beta \log \kappa(t)$) after

removing low-velocity outliers (< 5% peak speed). The absolute deviation between the observed β coefficient and the theoretical value of 1/3 serves as an index of relaxed motor planning; lower deviation indicates more natural, expert-like kinematic generation.

- Spatial and Temporal Consistency (The Accuracy of Movement)
  - Positional Consistency: To measure spatial accuracy independent of speed, we employed Dynamic Time Warping (DTW) on the coordinate trajectories (x, y). Unlike Euclidean distance, DTW accounts for temporal misalignments by non-linearly warping the time axis to find the optimal match between the novice's stroke and the expert reference. We computed DTW on arc-length resampled trajectories (to ensure spatial uniformity) using a Sakoe–Chiba band [31] constraint (10% of path length) to prevent pathological warping.
  - Speed-Accuracy Coupling: We evaluated motor efficiency by analyzing the relationship between spatial error (DTW distance) and peak velocity across repetitions. High-performance motor skill is characterized by the ability to maintain low spatial error even as momentum increases (breaking the standard Fitts' Law [9] trade-off observed in novices).
- Expressive Dynamics (The Style of Movement)
  - Velocity Consistency: We computed the Coefficient of Variation (CV) of stroke duration and peak velocity across the five repetitions of each primitive to quantify rhythmic stability.
  - Pressure Control: We computed the coefficient of variation (CV) of the normalized stylus pressure profile to assess the user's ability to maintain or intentionally modulate pressure stability.

- **Expressive Quality (Subjective):**
  - *Self-Assessment:* Participants rated their work on 5-point Likert scales regarding communicative intent (e.g., "I was able to convey my intended expression").
  - *Expert Ratings:* A panel of professional illustrators blind-rated anonymized drawings on a 10-point scale across four dimensions: originality, expressiveness, technical execution, and overall aesthetic quality.
- **User Experience:** Perceived helpfulness of feedback and prompts, and workload (NASA-TLX [12]).
- **Qualitative Insights:** Thematic analysis of interviews and think-aloud protocols during drawing, focusing on perceived challenges, learning strategies, and the role of feedback/prompts in expression.

7.8 Statistical Analyses

To assess the impact of the intervention on motor learning, we employed Linear Mixed-Effects Models (LMMs) for each performance metric (e.g., SPARC, DTW). LMMs were chosen for their robustness in handling repeated measures and their ability to account for inter-subject variability in baseline motor skills (random effects) while estimating the effects of the intervention (fixed effects). For each metric, we fitted a model with Group (4 levels) and Attempt Number (continuous) as fixed effects. We included the Group × Attempt Number interaction term to determine if the rate of learning differed significantly between the experimental and control groups. A significant interaction indicates that the intervention altered the trajectory of improvement over time.

To control for individual differences in initial skill level and variations in stroke complexity, we included crossed random intercepts for both Subject and Stroke Type. This formulation allows the model to account for distinct baselines across participants (e.g., prior artistic experience) and drawing primitives (e.g., spirals are inherently harder than lines), ensuring that our analysis isolates the relative improvement (learning) rather than absolute performance differences. The general model structure was defined as:

$$\begin{aligned}\text{Metric}_{ijk} = \beta_0 + \beta_1(\text{Group}) + \beta_2(\text{Attempt}) \\ + \beta_3(\text{Group} \times \text{Attempt}) + S_{0j} + P_{0k} + \epsilon_{ijk}\end{aligned}$$

where Metric_{ijk} is the score for subject j on stroke type k at attempt i. Here, S_{0j} represents the subject-specific random intercept, and P_{0k} represents the random intercept for the stroke primitive. Statistical significance was determined using Type III ANOVA (Satterthwaite's method) with a significance threshold of $\alpha = 0.05$

8 Results

We present results organized by the four measurement domains outlined in our methodology: technical proficiency (objective stroke data), expressive quality (subjective self-assessments and expert ratings), user experience (NASA-TLX), and qualitative insights (interviews and think-aloud).

8.1 Technical Proficiency (Objective)

We organized our objective analysis of motor skill acquisition into three components: Fluency (the quality of movement), Accuracy and Efficiency (fidelity to the target under speed), and Expressive Control (modulation of force and speed). We analyzed the impact of the feedback interventions on motor learning using LMMs with crossed random intercepts for Subject and Stroke Type. The results for smoothness (SPARC), accuracy (DTW), and naturalness (Speed/Power Law) are summarized in Table 4.

8.1.1 Motor Fluency and Smoothness. We first examined whether feedback enabled novices to transition from constrained, segmented movements to fluid, biologically natural motor planning.

- Smoothness (Spectral Arc Length, SPARC): We observed a significant Group × Attempt interaction for SPARC ($F(3, 2140.3) = 8.21, p < 0.05, \eta_p^2 = 0.15$), indicating that the rate of smoothness acquisition differed significantly between conditions. While the Control group (G1) showed a relatively flat learning curve, the Full Motus group (G4) exhibited an increasing trend, demonstrating rapid gains in movement smoothness. Post-hoc Tukey comparisons confirmed that by the final attempt, G4 achieved significantly higher SPARC scores than G1 ($p < 0.01$) and G2 ($p < 0.05$). Real-time multimodal feedback (visual overlays and descriptive text) effectively reduced movement jerkiness more than practice alone.
- Biological Naturalness (2/3 Power Law): We examined the R^2 fit to the 2/3 Power Law to determine if participants' movements converged toward natural biological dynamics (where

Metric	Fixed Effect	F-Value	df	p-value	Effect Size (η_p^2)
Smoothness (SPARC)	Group	3.22	[3, 32.3]	> 0.05	0.01
	Attempt (Trial)	**7.6**	**[1, 2140.1]**	**$< 0.05*$**	**0.07**
	Group × Attempt	**8.21**	**[3, 2140.3]**	**$< 0.05*$**	**0.15**
Naturalness (2/3 Power)	**Group**	**10.29**	**[3, 32.2]**	**$< 0.05*$**	**0.13**
	Attempt (Trial)	4.2	[1, 2140.1]	> 0.05	0.05
	Group × Attempt	**12.2**	**[3, 2140.6]**	**$< 0.05*$**	**0.12**
Accuracy (DTW)	**Group**	**7.47**	**[3, 32.4]**	**$< 0.05*$**	**0.08**
	Attempt (Trial)	**13.7**	**[1, 2140.3]**	**$< 0.005 **$**	**0.14**
	Group × Attempt	**9.57**	**[3, 2140.7]**	**$< 0.05*$**	**0.09**

Table 4: Results of LMMs for Technical Metrics for G1 to G4. Note: Significant effects ($p < 0.05$) are bolded. Interaction effects indicate differences in learning rates.

velocity scales with local curvature). The Linear Mixed-effects Model (LMM) yielded a significant Group × Attempt interaction ($F(3, 2140.6) = 12.2, p < 0.05$). While the Control group showed minimal change in power law adherence, suggesting they maintained a stiff, constrained drawing strategy, the Motor Feedback groups demonstrated a significant increase in fit. Post-hoc comparisons confirmed that G4 achieved significantly higher adherence than G1 by the final attempt ($p < 0.05$). This indicates that the real-time feedback helped participants transition from "tracing" behavior to more fluid, ballistic execution consistent with expert motor patterns.

8.1.2 Spatial and Temporal Consistency. Next, we assessed the fidelity of the final stroke trajectory relative to the intended expert path and the efficiency with which that fidelity was achieved.

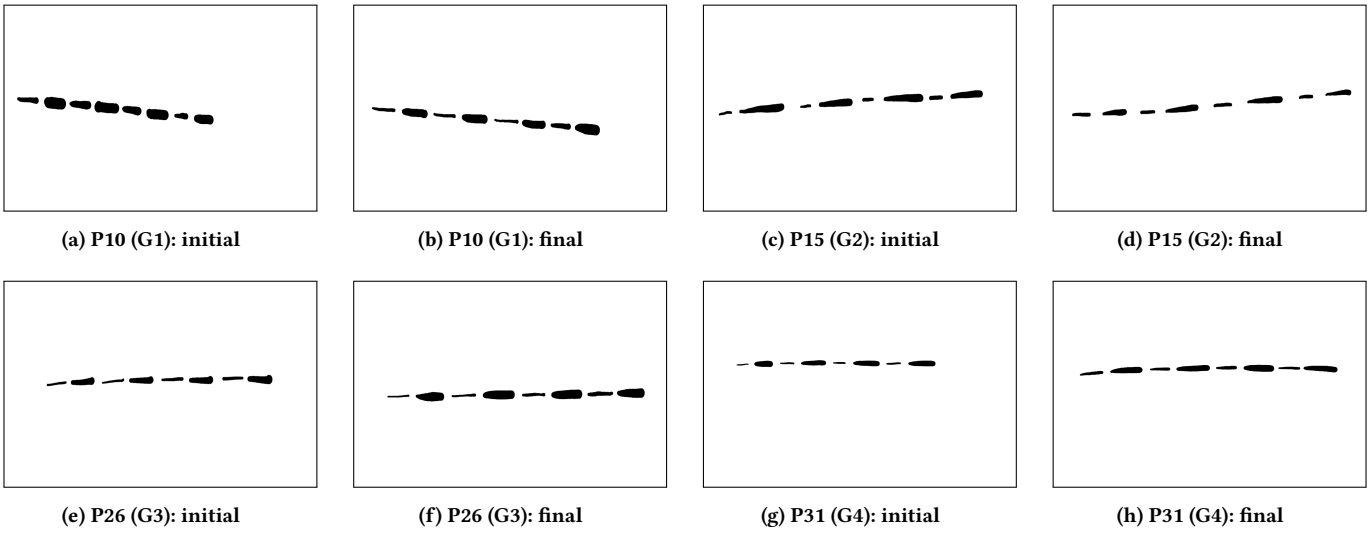
- Positional Consistency (DTW): Analysis of Dynamic Time Warping (DTW) distance revealed a significant main effect of Attempt ($F(1, 2140.3) = 13.7, p < 0.005$), confirming that all participants improved their shape replication accuracy over time. Additionally, the Group × Attempt interaction was also significant ($F(3, 2140.7) = 9.57, p < 0.05$). Post-hoc analysis showed that the motor-feedback groups (G3, G4) converged toward the target shape significantly faster than the Control group, resulting in lower final DTW distances (higher accuracy).
- Speed–Accuracy Coupling: To rule out the possibility that accuracy gains were simply a result of slower, more cautious execution (a speed-accuracy trade-off), we analyzed the relationship between execution speed and error rates. The LMM for Drawing Speed showed a significant main effect of Attempt ($F(1, 2140.4) = 14.1, p < 0.05$), indicating that all participants drew faster as they became familiar with the task. There was no significant negative interaction for speed ($p > 0.05$), meaning the Motor Feedback groups (G3, G4) did not slow down compared to controls. When viewed alongside the DTW results, where G4 significantly outperformed G1, this confirms a shift in the speed-accuracy operating point: participants in the Full Motus condition achieved higher accuracy at equal or higher speeds, demonstrating a genuine improvement in motor efficiency.

8.1.3 Expressive Dynamics and Control. We analyzed the intentional control over physical input signals, including the parameters previously identified as novice challenges: pressure, velocity, and tilt consistency. The descriptive statistics (mean and SD) for groups G1-G4 are tabulated in Table 5.

- Parameter Convergence: We observed that numerous strokes exhibited substantial improvement across the five attempts. To formally assess this effect, we conducted a Mann-Whitney U test together with Levene's test on the expert strokes (N=9) and on the novice groups G3 and G4 who received motor feedback (N=18) for first attempt and last attempt, as the data violated normality assumptions. An analysis of Parallel Consistency indicates an improvement in pressure dynamics, moving from $U = 130$, $\mathbf{p < 0.05}$, $W(1, 25) = 0.989$, $p > 0.05$ to $U = 121$, $\mathbf{p > 0.05}$, $W(1, 25) = 0.568$, $p > 0.05$. These statistics indicate that although the median pressure consistency of novices differed significantly from that of experts during the first attempt, it had converged to match expert-level performance by the final attempt, while the variance of consistency remained aligned throughout. The same effect can be observed for Temporal Spatial Spiral where the initial statistics are $U = 45$, $\mathbf{p < 0.05}$, $W(1, 25) = 0.227$, $p > 0.05$ and final statistics are $U = 49$, $\mathbf{p > 0.05}$, $W(1, 25) = 2.423$, $p > 0.05$. This behavior is common in velocity and tilt as well. For Pressure Crescendo the first attempt's median velocity statistics suggest a inconsistency with the experts $U = 135$, $\mathbf{p < 0.05}$, $W(1, 25) = 0.104$, $p > 0.05$ which aligned with experts at final attempt $U = 122$, $\mathbf{p > 0.05}$, $W(1, 25) = 0.184$, $p > 0.05$. The median tilt consistency of the Temporal Spatial Spiral improved from $U = 40$, $\mathbf{p < 0.05}$, $W(1, 25) = 6.137$, $p < 0.05$ to $U = 89$, $\mathbf{p > 0.05}$, $W(1, 25) = 4.411$, $p < 0.05$ despite the variance did not improve.
- Motor Alignment Across Attempts: Although most strokes shift from inconsistency to consistency with the expert profile over the course of motor training, a few continue to exhibit discrepancies. Nevertheless, the analysis of Pearson correlations across the five attempts demonstrates a clear trend toward convergence with the expert patterns. For instance, the variance of pressure consistency for Confident Curve remains significantly different from the expert even at attempt 5 [$W(1, 25) = 15.058$, $p < 0.05$]; however,

Parameter	Group 1				Group 2				Group 3				Group 4				Group 5			
	Attempt 1		Attempt 5		Attempt 1		Attempt 5		Attempt 1		Attempt 5		Attempt 1		Attempt 5		Attempt 1		Attempt 5	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
Pressure consistency	0.147	0.055	0.170	0.076	0.173	0.040	0.178	0.054	0.149	0.062	0.147	0.055	0.128	0.034	0.129	0.035	0.151	0.027	0.164	0.066
Velocity consistency	0.231	0.097	0.286	0.125	0.231	0.118	0.229	0.100	0.228	0.093	0.236	0.108	0.147	0.078	0.246	0.123	0.121	0.057	0.159	0.108
Tilt consistency	2.043	1.441	2.528	1.918	2.339	1.393	2.679	1.799	1.816	1.225	1.573	1.069	1.263	0.480	1.460	0.542	1.641	1.685	1.233	0.879

Table 5: Example Descriptive statistics of Inconsistency for Pressure Crescendo across the Evaluation Groups (G1–G4) and the Validation Group (G5).



(a) P10 (G1): initial **(b) P10 (G1): final** **(c) P15 (G2): initial** **(d) P15 (G2): final**

(e) P26 (G3): initial **(f) P26 (G3): final** **(g) P31 (G4): initial** **(h) P31 (G4): final**

Figure 5: Drawings of Pressure Staccato by participants P10, P15, P26, and P31 from Groups G1–G4, respectively. Notably, participants in G3 and G4 exhibited clear improvement between their initial and final attempts, demonstrating the impact of feedback and combined interventions on stroke refinement.

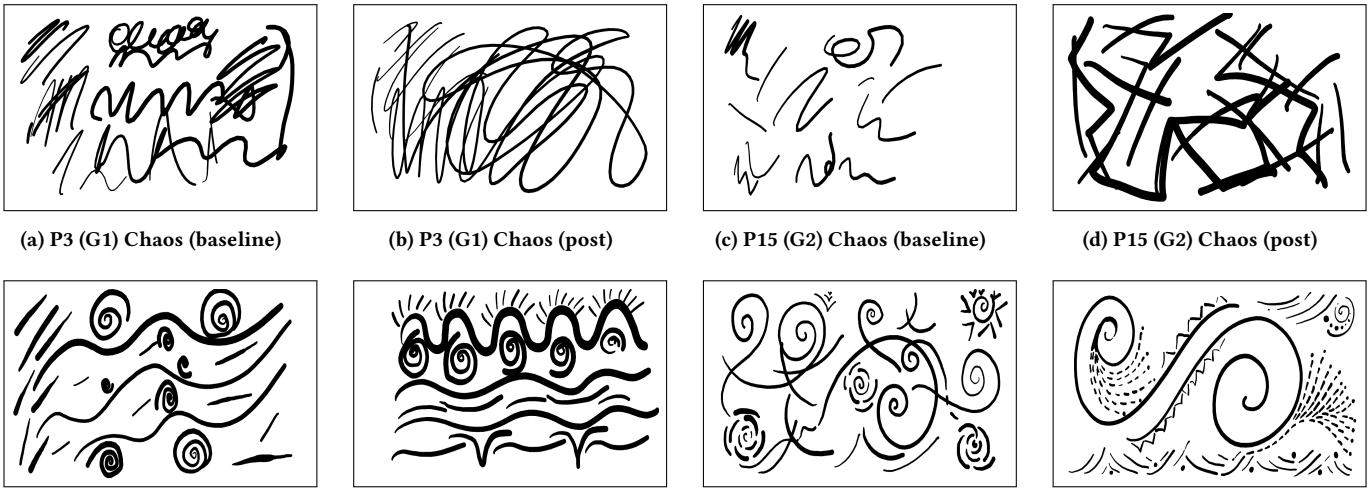
the Pearson correlation steadily improves from $r = 0.963$ at attempt 1 to $r = 0.984$ at attempt 5. Furthermore, the median of tilt inconsistency for Responsive Curves was significantly different from the expert profile, as indicated by $U = 137$, $p < 0.05$. However, the corresponding Pearson correlation improved from $r = 0.343$ to $r = 0.497$, reflecting gradual convergence toward the expert pattern.

Fig. 5 illustrates the progression of novice performance on Pressure Staccato, showing the first and final attempts from one representative participant in each group. The contrast highlights how different conditions shaped motor improvement. In the control group (G1), little change is visible across attempts: pressure remains uneven and the line spacing inconsistent and not horizontal. With ideation prompts alone (G2), participants showed greater intentionality but still struggled to maintain consistent alternation of pressure. By contrast, participants who received motor feedback (G3) demonstrated clearer structural gains, with horizontal alignment and more regular pressure switching evident in the final attempt. The full system (G4) yielded the most substantial improvement, combining horizontal stability with deliberate, rhythmic pressure modulation across marks. Together, these visual examples underscore how feedback, particularly when paired with prompts, helped novices internalize the expressive pattern and execute the stroke with greater fluency.

8.2 Perceived Expressiveness (Subjective and Expert)

Perceived expressiveness refers to the extent to which participants felt their drawings successfully communicated intended emotions and expressive qualities. This was measured in two complementary ways: (i) participant self-assessments on Likert scales, rating how expressive, emotionally satisfying, and communicative their drawings were; and (ii) expert evaluations of pre- and post-intervention drawings, using a 10-point scale across dimensions of motor/technical skill, expressive ability, and overall artistic quality.

8.2.1 Expert Evaluation. To assess how participants' work was perceived by professional evaluators, we asked five expert illustrators to rate all pre- and post-intervention drawings from Groups 1–4 (in Fig. 6 we illustrate the pre- and post-intervention creative task output images for randomly chosen participants). Experts provided evaluations on a 10-point Likert scale across seven dimensions, which we organized into three categories: motor/technical skill, expressive ability, and overall artistic quality. Inter-rater reliability was high across experts (cronbach's $\alpha = 0.88$), indicating strong consistency in judgments. To ensure objectivity, the evaluation was performed by blind expert raters who were unaware of the experimental conditions.



(a) P3 (G1) Chaos (baseline) (b) P3 (G1) Chaos (post) (c) P15 (G2) Chaos (baseline) (d) P15 (G2) Chaos (post)

(e) P25 (G3) Calm (baseline) (f) P25 (G3) Calm (post) (g) P37 (G4) Happiness(baseline) (h) P37 (G4) Happiness (post)

Figure 6: Pre- and Post-intervention abstract drawing samples from participants P3, P15, P25, and P37 from each control/test group.

Row	Pre			Post		
	Expressive	Motor	Overall	Expressive	Motor	Overall
G1	4.87 (0.95)	5.45 (1.15)	4.9 (1.08)	4.83 (1.34)	5.30 (0.80)	4.93 (1.25)
G2	4.5 (1.29)	4.56 (1.32)	4.62 (1.28)	5.61(1.19)*	5.57 (0.77)	5.56 (1.01)
G3	5.39 (0.96)	5.00 (0.36)	5.12 (0.68)	5.96 (1.19)	5.99 (0.96)*	5.86 (1.10)
G4	5.18 (0.36)	5.2 (1.04)	5.2 (1.20)	5.85 (1.54)	6.00 (1.44)	5.95 (0.55)
G5	4.90 (0.38)	4.80 (0.65)	4.60 (1.25)	5.91 (0.76)	6.02 (1.14)*	5.89 (1.09)*

Table 6: Mean and SD expert assessment scores at pre-test and post-test across experimental groups G1 through G5. Asterisks indicate a statistically significant difference between pre- and post-test scores ($*p < 0.05$)

We first conducted Shapiro–Wilk tests to examine normality within each group–category comparison; all data were normally distributed ($p > 0.05$). We then ran paired-sample t-tests comparing pre- and post-intervention scores within each group (the mean pre- and post expert evaluation scores for G1 - G4 are tabulated in Table 6.

- Motor Skill Gain: Participants in Group 3 (Feedback Only) showed reliable gains from pre- ($M = 5.00$, $SD = 0.36$ to post- ($M = 5.99$, $SD = 0.96$) intervention ($t(8) = -2.23, p < 0.05$), suggesting that multimodal motor feedback was particularly effective at improving the technical execution of strokes.
- Expressive Ability: Participants in Group 2 (Prompt Only) demonstrated significant improvement (pre: $M = 4.50, SD = 1.29$, post: $M = 5.61, SD = 1.19$) ($t(8) = -3.10, p < 0.05$), indicating that embodied prompts supported clearer communication of affective intent.
- Overall Artistic Quality: We pooled participants from all guided conditions (G2–G4) and found robust improvements when compared to their baseline work (pre: $M = 4.97, SD = 1.19$, post: $M = 5.79, SD = 1.45$) ($t(26) = -3.54, p < 0.001$). This finding highlights that even minimal guidance, whether through prompts, feedback, or both, enhanced the perceived quality of novice outputs.

8.2.2 Self-Assessment and Perceived Success. Participants' own reports echoed the patterns observed in expert evaluations. When asked to rate their work on expressiveness, emotional quality, and communicative clarity, groups receiving structured support reported higher perceived success than those with limited or no guidance. Mean ratings (on a 7-point scale) were lowest for G3 ($M = 3.1, 3.0, 3.2$), somewhat higher for G2 ($M = 3.5, 3.4, 3.2$), and highest for G4, the full Motus condition ($M = 3.89, 4.11, 3.78$). These results suggest that while ideation prompts alone (G2) enhanced participants' confidence relative to feedback alone (G3), the combined intervention (G4) produced the strongest subjective gains across all dimensions. Participants in G4 consistently rated their drawings as more expressive, emotionally satisfying, and effective at conveying intended feelings, indicating that motor feedback

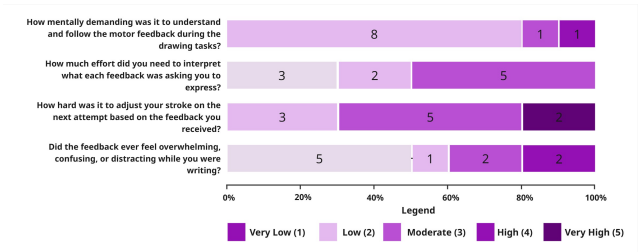


Figure 7: Horizontal bar chart showing the 5-point Likert scale ratings from G3.

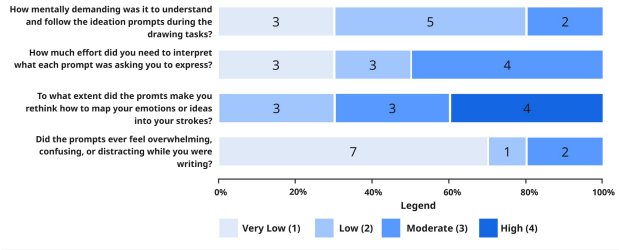


Figure 8: Horizontal bar chart showing the 5-point Likert scale ratings from G2.

and embodied prompts acted synergistically to bolster creative self-efficacy.

8.2.3 The Motor-Expressive Link: Mediation Analysis. To test the core hypothesis that objective motor fluency mediates expressive quality, we conducted a simple mediation analysis (PROCESS model 4).

We defined a composite mediator representing Objective Motor Gain based on the Δ gain (Post - Pre) across three key objective metrics (SPARC, DTW Distance, and the 2/3 Power Law Deviation). The dependent variable was the change in Expert-Rated Expressive Ability (Post-Intervention minus Pre-Intervention).

The indirect effect of the Full Motus Intervention (X) on the Expert Expressive Gain (Y) via the Objective Motor Gain (M) was significant ($\beta_{indirect} = 0.32$, 95% CI $[0.14, 0.52]$). This finding provides systematic evidence that the observed enhancement in novices' expressive ability is significantly channeled through the acquisition of quantifiable motor skills (fluency and control), supporting the core design paradigm of motor-mediated creativity.

8.3 Cognitive Load

Motor Feedback. Mean cognitive load ratings varied across different aspects of the motor feedback drawing tasks (Fig. 7). Understanding and following motor feedback during drawing tasks received a relatively low mean rating ($M = 2.30, SD = 0.67$), indicating low to moderate cognitive demand. Interpreting feedback content required similar effort ($M = 2.20, SD = 0.92$), while adjusting stroke execution based on received feedback was more demanding ($M = 3.10, SD = 1.10$). Participants reported low levels of feeling overwhelmed, confused, or distracted by the feedback ($M = 2.10, SD = 1.29$).

Overall, the cognitive load was generally manageable across most aspects of the motor feedback tasks, with stroke adjustment being the most cognitively demanding component. The low ratings for feeling overwhelmed suggest that participants could effectively process and utilize the motor feedback without significant interference during the drawing tasks.

Ideation Guidance. Mean cognitive load ratings revealed low demands for most ideation-related aspects of the drawing tasks (Fig. 8). Understanding and following ideation prompts required minimal cognitive effort ($M = 1.90$, $SD = 0.74$), as did interpreting what each prompt was asking participants to express ($M = 2.10$, $SD = 0.88$). Participants also reported low levels of feeling overwhelmed, confused, or distracted by the prompts ($M = 1.50$, $SD = 0.85$).

Notably, prompts were effective in encouraging participants to rethink their approach to emotion–stroke mapping ($M = 3.10$, $SD = 0.88$), indicating that while the cognitive burden remained low, the prompts successfully encouraged reflection on creative expression strategies. Overall, the ideation prompts were cognitively undemanding yet effective, achieving their intended purpose of stimulating creative reflection without imposing significant mental effort or causing confusion.

Linking Cognitive Load and Performance. To assess if cognitive effort influenced outcomes, we correlated NASA–TLX ratings with expert-rated motor quality. Results showed that lower cognitive load scores on "interpreting feedback" were moderately associated with higher motor control gains ($r = -0.41$, $p < 0.05$), suggesting that when participants found the feedback easy to process, they improved more in line quality and pressure control. This finding indicates that the system's multimodal design was effective not only because it provided actionable guidance but because it minimized extraneous cognitive demands, allowing participants to focus their mental effort on motor refinement.

8.3.1 Qualitative Feedback. We conducted a thematic analysis of semi-structured interviews. Three overarching themes emerged:

- Uncertainty Without Feedback. Participants in G1 and G2 often expressed uncertainty about their strokes, with comments such as "I couldn't tell what I was doing wrong" or "I just kept drawing until something looked okay." This aligns with objective measures showing higher variance in pressure and velocity for non-feedback groups.
- Feedback as Diagnostic and Corrective. Participants in G3 and G4 described feedback as a "mirror" that revealed specific breakdowns in motor execution. The radar plots were valued for surfacing which parameter (e.g., pressure or velocity) needed attention, while temporal overlays were cited as "pinpointing exactly where my stroke went off." One participant remarked, "I didn't realize I was lifting pressure too early until the system showed me." However, some in G3 noted feeling "overloaded" by multiple feedback channels without a creative outlet to apply them.
- Prompts as Creative Triggers. Embodied prompts were consistently described as sparking imaginative strategies. Participants noted that prompts like "draw as if stretching after waking up"

allowed them to "feel the motion first, then translate it to the stroke." Importantly, the exploratory hints (e.g., continuous vs. fragmented lines) were seen as a gentle structure that "got me started" without constraining creativity. Several G4 participants emphasized the synergy between feedback and prompts: "The feedback showed me how to control my lines, and the prompt gave me a reason to use that control."

Overall, qualitative findings reinforce quantitative results: feedback supports technical mastery, prompts stimulate ideation, and their combination provides the richest experience for novices.

9 Robustness and Generalization (Cross-Device Validation)

To ensure the robustness of our conclusions against external factors (hardware specificity) and internal validity concerns (potential carry-over effects from the main study's cohort), we conducted a focused validation study with an independent sample using an alternate, widely used hardware platform.

9.1 Validation Study Design and Data Handling

The validation study (Group 5 or G5) addressed three main goals: (i) generalizing effects beyond the Wacom ecosystem, (ii) replicating findings with an independent sample ($N = 9$), and (iii) examining the moderating roles of individual differences.

Participants: Nine right-handed novices were recruited who had not participated in any prior phase. This ensures that observed gains are not attributable to carry-over effects from the formative study.

Apparatus and Sampling Differences: The validation study utilized an iPad Pro (12.9", 4th gen) with an Apple Pencil (2nd gen). The data processing pipeline remained identical to that described in Section 5.4, with the exception that data were collected at a sampling rate of 240Hz. Additionally, as the specific pressure quantization levels are not disclosed by the manufacturer, we utilized the normalized continuous pressure values (range 0–1) provided by the operating system.

Experimental Design: We instantiated the Full Motus intervention (Group 4 equivalent) to test whether the integrated effect generalizes. We specifically measured prior art experience, baseline fine-motor ability, and body self-awareness to use as covariates in confirmatory models.

9.2 Validation Results: Replication and Consistency

9.2.1 Motor Metric Replication: Mixed-effects models on pooled data (Wacom + iPad) revealed significant Time effects (Pre- to Post-intervention) for all primary motor metrics (all $p < 0.01$). Further, there were no statistically significant Time × Device interactions (all $p > 0.05$). This demonstrates that the Motus intervention's effectiveness generalizes across hardware platforms.

The descriptive statistics for the three motor parameters of G5 participants is tabulated in Table 5. We also examined whether there were significant differences in consistency at attempt 5 of G4 and G5 across studies to verify whether there is a carryover effect. Given that the data were not normally distributed according to the Shapiro-Wilk test, we employed the Mann-Whitney U test to assess differences in medians and Levene's test to assess differences in variances. For both pressure and tilt across all strokes, the results of the Mann-Whitney U and Levene's tests indicated no significant differences ($p > 0.001$). In contrast, for velocity, a significant difference was observed only for Pressure Staccato ($p < 0.05$).

Consistent with the findings of the evaluation study, the motor training sessions assisted novice users in bringing their motor parameters closer to expert-level performance. For strokes like Pressure Crescendo, where precise pressure modulation is essential, novices initially exhibited substantially higher inconsistency relative to experts, with $U = 15$, $\mathbf{p < 0.05}$, $W(1, 16) = 0.013$, $p > 0.05$. By the fifth attempt, however, this discrepancy had diminished, improving to $U = 22$, $\mathbf{p > 0.05}$, $W(1, 16) = 2.071$, $p > 0.05$, indicating convergence toward expert performance (i.e., the median inconsistency aligned with experts over the motor training while the variance remained comparable). A similar pattern is observed for Accelerando Line in terms of velocity, where novice performance improved from $U = 72$, $\mathbf{p < 0.05}$, $W(1, 16) = 1.888$, $p > 0.05$ to $U = 63$, $\mathbf{p > 0.05}$, $W(1, 16) = 0.952$, $p > 0.05$.

For strokes that did not exhibit convergence toward expert performance even by the final attempt that users did, we assessed improvement by comparing the Pearson correlation between the first and final attempts. For pressure in Rhythmic Pulses and Parallel Consistency, the correlation increased from $r = 0.874$ to $r = 0.902$ and from $r = 0.507$ to $r = 0.605$, respectively. Similarly, for Confident Curve and Rhythmic Wave, where velocity modulation is the primary factor, the velocity correlation improved from $r = 0.661$ to $r = 0.723$ and from $r = 0.395$ to $r = 0.594$, respectively.

9.2.2 Expert Evaluations. To verify that the qualitative gains observed in the main study were not artifacts of the specific hardware (Wacom Movink) or the initial cohort, we replicated the expert evaluation protocol with the Validation Group (G5, iPad users).

Method. The same panel of five blind expert raters evaluated the baseline (Pre) and final (Post) creative drawings of the G5 participants. Ratings were provided on the same 10-point scale across Technical Execution, Expressive Ability, and Overall Quality. Inter-rater reliability remained high (Cronbach's $\alpha = 0.86$). Data were normally distributed (Shapiro-Wilk $p > 0.05$), allowing for paired-sample t-tests.

Results. The analysis confirmed that the Full Motus intervention yielded significant improvements in creative quality, independent of the hardware platform. The corresponding pre vs. post mean expert evaluation scores for G5 are tabulated in Table 6.

- Technical Execution: Participants demonstrated a significant increase in motor control scores from baseline ($M = 4.8, SD = 0.65$) to post-intervention ($M = 6.02, SD = 1.14$), ($t(8) = -2.23, p < 0.05$). This mirrors the gains seen in Group 4, confirming that the feedback mechanisms are effective on standard consumer tablets.
- Expressive Ability: Ratings for the ability to convey the intended emotion (e.g., Chaos, Calm) improved significantly ($t(8) = -2.67$, $p < 0.01$), suggesting that embodied prompts remain effective triggers for expression even on different devices.
- Overall Artistic Quality: Most notably, the holistic quality of the artworks improved from a mean of 4.6 to 5.89 ($p < 0.01$).

Comparison with Main Study. To ensure consistency, we compared the magnitude of improvement (Post-Pre delta) between the Main Study Full Motus group (G4) and the Validation group (G5) using an independent samples t-test. We found no significant difference in the learning gains between groups for any dimension ($p > 0.05$). This strongly suggests that the efficacy of Motor-Mediated Creativity is robust across different hardware ecosystems and user cohorts.

10 Discussion and Future Work

Our findings demonstrate that explicit motor training, when paired with embodied ideation prompts, significantly enhances both the technical fluency and expressive quality of novice digital drawings. Beyond these empirical results, this work offers broader theoretical implications for the design of Creativity Support Tools (CSTs).

10.1 Implications for Theory: From Automation to Embodied Scaffolding

The dominant paradigm in CST research has largely focused on "lowering the threshold" for expression by automating low-level execution (e.g., stroke beautification, predictive sketching). While this approach democratizes output, it risks creating an "expressive gap" where users produce high-fidelity results without developing the embodied agency to control them. Our work challenges this automation-centric view by proposing Motor-Mediated Creativity, a theoretical perspective that positions motor fluency not as a barrier to be removed, but as a generative resource to be cultivated.

This shift advances our understanding of CSTs in two key ways. First, it demonstrates that motor learning and creative ideation are coupled processes. The significant interaction effects in our study suggest that novices do not simply "have an idea" and then "execute it"; rather, the ability to execute (motor fluency) shapes the kinds of ideas they can entertain. By scaffolding motor skills, we are effectively expanding their "expressive vocabulary," enabling them to form intentions that were previously inaccessible. Second, our results validate the role of embodied simulation in digital tool design. The success of our embodied prompts confirms that users can leverage sensorimotor metaphors (e.g., "bursting," "heavy") to bridge the gap between abstract affect and kinematic execution. This suggests that future CSTs should move beyond symbolic instruction (e.g., "draw a curved line") toward prompts that trigger the kinesthetic systems underlying expression.

10.2 Limitations and Future Work

While promising, our findings are bounded by several constraints. First, our stroke taxonomy is illustrative rather than exhaustive; while it covers key primitives (pressure, rhythm), it does not capture the full diversity of artistic mark-making. Future work should expand this library in collaboration with artists from varied traditions. Second, our study focused on short-term acquisition. Longitudinal research is needed to determine if these motor gains persist over time or transfer to complex, unguided creative workflows. Finally, our participant pool was restricted to novices. Expert artists may require different feedback modalities that support stylistic refinement rather than foundational control. Addressing these limitations will be crucial for scaling motor-mediated creativity from a training intervention to a pervasive interaction paradigm.

Investigating Motivational Mechanisms: While this study primarily evaluated skill acquisition, the system's design, which converts ambiguous errors into actionable metrics, aligns with theoretical frameworks for intrinsic motivation and Creative Self-Efficacy. By providing clear goals and immediate feedback, such tools have the potential to sustain engagement through the "zone of proximal development," preventing the frustration that often leads novices to abandon digital drawing. Future research should explicitly operationalize this by measuring learner persistence and motivation using instruments like the Intrinsic Motivation Inventory (IMI) over longitudinal practice.

11 Conclusion

This paper introduced Motus, an interactive environment that foregrounds motor-mediated creativity in digital drawing. Through a formative study, we identified the specific challenges novices face in controlling pressure, velocity, and directionality, and demonstrated how feedback and prompts can make motor–expression relationships more interpretable. Building on these insights, we designed and evaluated a system that combines multimodal feedback with embodied ideation prompts, showing that both elements independently and jointly enhance expressive outcomes.

Our findings contribute to HCI by reframing creativity support around embodied skill acquisition rather than feature expansion. By treating motor execution not as a barrier but as a site of design intervention, we demonstrate that explicit training of expressive primitives can meaningfully improve both technical fluency and creative expression. While our evaluation focused on a constrained set of strokes and short-term learning, the results provide the systematic evidence that motor training can be productively integrated into digital creativity systems.

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Appendix

A Stroke Taxonomy

This section introduces the taxonomy of the twelve primitive strokes that form the foundation of our study. The taxonomy organizes strokes into pedagogical blocks and articulates both their expressive transfer potential and instructional motives, serving as the conceptual backbone for our training and evaluation framework. In Table. 7 we tabulate the pedagogical and transfer intent of the 12 primitive strokes we have considered.

B Expert Data Collection

In this section, we outline the procedures for gathering insights from professional illustrators. The expert interviews were designed to capture perspectives on stroke naturalness, pedagogical sequencing, expressive potential, and overall reflections, providing qualitative depth to our stroke taxonomy and training design.

- **Naturalness and Intuitiveness:**
  Which 2-3 strokes felt most natural or intuitive to you?
- **Beginner Challenges:**
  - Which would you teach first to someone new to digital drawing?
  - Which strokes did you use to convey specific emotions or ideas in your artwork?
  - Were there moments where a particular stroke technique helped you express something you couldn't before?
  - Would you approach a similar creative task differently now compared to before learning these strokes?
- **Teaching for Expression:**
  - Which strokes do you believe offer the greatest potential for artistic expression?
  - Were there strokes that seemed particularly good for conveying emotion or mood?

Block	Stroke	Expressive Transfer	Pedagogical Motive
Pressure Dynamics	Pressure Crescendo	Emotional intensity gradation, emphasis control in mark-making; ability to create dramatic builds and releases in visual compositions.	Builds touch sensitivity and pressure awareness; establishes foundation for expressive line variation without relying on speed changes.
Pressure Dynamics	Pressure Staccato	Rhythmic emphasis patterns, textural variation control; creates visual beats and accents in compositional work.	Develops pressure memory and rapid motor switching; trains independent control of pressure from other mark parameters.
Pressure Dynamics	Pressure Sustain	Bold statement making, unwavering confidence expression; creates strong foundational elements in compositions.	Builds sustained grip strength and endurance; teaches separation of pressure control from directional control.
Temporal Dynamics	Accelerando Line	Dynamic energy building, momentum creation; ability to inject increasing energy and urgency into mark-making.	Develops kinesthetic awareness of speed changes; teaches temporal control as an expressive tool independent of pressure.
Temporal Dynamics	Rhythmic Pulses	Musical rhythm translation to visual marks; creates flowing, dance-like qualities in linear work.	Establishes internal timing mechanisms; connects temporal awareness to spatial execution for fluid mark-making.
Directional Confidence	Confident Curve	Graceful movement quality, organic form creation; ability to create sophisticated curved forms that feel alive and natural.	Builds confidence in complex trajectories; develops trust in motor planning for sophisticated curved paths.
Directional Confidence	Angular Precision	Geometric precision, structural authority; creates architectural and mechanical drawing foundations.	Develops angular accuracy and sharp directional control; builds precision in geometric mark-making.
Spatial Relationships	Parallel Consistency	Structural foundation building, architectural line control; creates ordered, systematic mark relationships.	Develops spatial measurement skills; builds foundation for complex compositional relationships and systematic mark-making.
Spatial Relationships	Responsive Curves	Conversational mark-making, compositional dialogue; creates interactive relationships between elements.	Develops compositional thinking; teaches marks as part of larger visual conversations rather than isolated elements.
Multi-Modal Integration	Directional Pressure Shift	Dimensional illusion creation, atmospheric depth; combines spatial direction with pressure variation for sophisticated effects.	Integrates pressure and directional skills; prepares for complex mark-making requiring multiple simultaneous controls.
Multi-Modal Integration	Temporal Spatial Spiral	Organic growth patterns, energy dissipation visualization; creates natural movement from intense center to calm periphery.	Integrates temporal and spatial complexity; develops coordination of speed and space for advanced expressive control.
Multi-Modal Integration	Rhythmic Wave	Musical visualization, harmonic rhythm translation; creates sophisticated relationships between multiple expressive elements.	Synthesizes all motor skills into coordinated multi-parameter control for advanced artistic expression.

Table 7: Stroke taxonomy: Twelve expressive stroke primitives organized into five pedagogical blocks, with corresponding expressive transfer goals and pedagogical motives.

- **Overall Reflection:**
  - Any surprises about how these strokes worked or felt?
  - Quick thoughts on combining these techniques effectively?

C Formative Study: Detailed Results

C.1 Motor Fluency

At this finer granularity, clear patterns have been observed. For seven of the twelve strokes, at least one parameter showed significantly higher variance among novices. For instance, in Pressure Crescendo, where sustained modulation of pressure is key, novices differed significantly from experts in pressure variance, $W(1, 49) = 4.490, , p < 0.05$ (Fig.2a). In Confident Curve which requires precise directional control, both pressure $W(1, 49) = 4.548, , p < 0.05$ and velocity $W(1, 49) = 5.286, , p < 0.05$ were significantly more variable for novices (Fig.2b). Similarly, Angular Precision revealed novice difficulty in maintaining steady velocity, $W(1, 49) = 4.748, , p < 0.05$ (Fig. 2c).

These challenges extended to tasks requiring relational control. For Parallel Consistency novices exhibited significantly higher variance in velocity, $W(1, 49) = 6.333, , p < 0.05$, as they attempted to maintain equal spacing between strokes (Fig.2d). A similar effect was found for Responsive Curves $W(1, 49) = 4.190, , p < 0.05$ (Fig.2e), where novices struggled to adapt one curve relative to another. Complex integrative tasks also proved difficult: in Temporal Spatial Spiral novices showed greater tilt variability, $W(1, 49) = 5.143, , p < 0.05$ (Fig. 2f), while in the Rhythmic Wave velocity variance again distinguished novices from experts, $W(1, 49) = 5.724, , p < 0.05$.

Taken together, these results demonstrate that novice challenges are not uniform but clustered around particular parameter–stroke combinations. Velocity control emerged as the most persistent difficulty, especially in strokes that demanded spatial precision (e.g., angular precision, parallel consistency). Pressure modulation was most problematic in strokes emphasizing gradual force control, while tilt variability became salient in more complex, multi-dimensional gestures. Importantly, the consistency of these findings across structurally similar strokes suggests that novice challenges are systematic rather than random, offering clear design targets for feedback and training interventions.

C.2 Qualitative Results

Participant reflections supported the above finding:

> *"The pressure control and the velocity control are giving really hard times, it is hard to be aware what speed and pressure I was using at every second."* — P2
>
> *"Pressure control, because it's hard to control muscle to draw smoothly."* — P6
>
> *"Managing the stroke width and keeping it straight at the same time, drawing parallel lines is challenging for me."* — P12

Challenge 2: Difficulty Linking Motor Actions to Intended Expression. Many novices reported struggling to translate their creative intentions into motor actions. While they could describe what they wanted a stroke to "feel like," they often did not know how to modulate velocity, pressure, or curvature to achieve it. Qualitative feedback illustrates this gap:

> *"I was not able to control precisely that I can gradually change my pressure and velocity (especially for the curves). It is hard to coordinate both pressure and velocity at the same time."* — P2
>
> *"Keeping the consistency while repeating the same shape is very challenging as I need to keep the pressure and other parameters consistent."* — P5
>
> *"During the pressure control, especially when asked to maintain pressure, I found moving the whole arm to draw the stroke a lot more consistent than just using the wrist. The same strategy applied for being asked to draw a line in the intended direction."* — P20

These reflections highlight that novices lack an embodied understanding of how motor parameters map onto expressive intentions.

Challenge 3: Managing Competing Demands Between Technique and Expression. Novices also struggled to simultaneously maintain technical control while considering expressive goals. Many reported difficulty focusing on both, leading to what could be described as divided attention:

> *"Controlling the pen with the amount of pressure, tilt and velocity directly reflects the artistic expression."* — P16
>
> *"Thickness and speed of the stroke can express strong or weak expression."* — P19
>
> *"I just learnt today that pressure, velocity matters in the world of digital drawing, which can help in expression such as drawing harder and faster when anger is supposed to be portrayed."* — P20
>
> *"The insight I got was how different emotions can also translate to how my hands were moving the pen, for example for prompts that reflect a calm emotion, I had to bring myself to that head space and kept my pen steady. Conversely, when the prompt was tension, there was an energy I brought to my hands, which made my movements on the tablet more erratic and random."* — P26

This challenge reflects the cognitive burden of coordinating multiple motor and conceptual demands in a digital drawing context, even without formal workload measurements.

D Formative Experiment: Post-study Questionnaire

This section details the post-study questionnaire administered during the formative experiment. The questionnaire assessed participant challenges, perceived effectiveness of feedback and prompts, and their support for creativity, learning, and motor expression. These responses inform refinements to both feedback design and experimental methodology.

- Rate how challenging each area was during the session: (1 = Not challenging at all, 5 = Extremely challenging)
  - Pressure Control: Varying stylus pressure for different line weights

  - Velocity Regulation: Controlling stroke speed for different effects
  - Directional Precision: Drawing smooth curves and maintaining intended directions
  - Stroke Timing: Coordinating timing of different parts of a stroke
  - Hand-Eye Coordination: Coordinating hand movements with screen feedback
- Feedback effectiveness – the following questions were repeatedly asked for each specific motor feedback type (1 = Not at all, 5 = Extremely):
  - Helpfulness: How helpful was this feedback for improving your strokes?
  - Intuitiveness: How easy was this feedback to understand?
  - Learning Support: How well did this feedback support your learning?
- Prompt Effectiveness – the following questions were repeatedly asked for each specific ideation prompt type (1 = Not at all, 5 = Extremely):
  - Creativity Support: How well did this condition support creative expression?
  - Idea Generation: How well did this help you generate drawing ideas?
  - Motor Expression: How well did this help you express ideas through movement?

E Evaluation Study Details

This section provides a comprehensive account of the evaluation study, including expert ratings, participant self-reports, and workload assessments. These measures triangulate how training interventions shaped both technical proficiency and expressive outcomes.

E.1 Expert Evaluation of Expressiveness

Here we describe how expert illustrators evaluated participants' pre- and post-intervention drawings. Ratings were collected on multiple dimensions of motor skill, expressive ability, and overall artistic merit, providing an external benchmark of progress beyond participant self-perception (on a 10-point Likert scale).

- **Motor/Technical skill**
  - Line quality:
    Scale: 1 (Very Poor) - 10 (Excellent)
    * 1-2: Lines are shaky, inconsistent, or poorly controlled
    * 3-4: Lines show some control issues, but basic intent is visible
    * 5-6: Lines are adequately controlled with minor inconsistencies
    * 7-8: Lines show good control and intentionality
    * 9-10: Lines demonstrate excellent control, confidence, and precision
  - Stroke consistency:
    Rate how consistently the artist applied pressure, speed, and direction across different strokes.
    Scale: 1 (Very Inconsistent) - 10 (Highly Consistent)
  - Motor control proficiency:
    Rate the artist's overall stylus control and hand-eye coordination as demonstrated in this work.
    Scale: 1 (Beginner/Poor Control) - 10 (Advanced/Excellent Control)
- **Expressive ability**
  - Emotional clarity:
    How clearly does this artwork communicate an emotional state or feeling?
    Scale: 1 (No Clear Emotion) - 10 (Very Clear Emotional Expression)
    * 1-2: No discernible emotional content
    * 3-4: Vague suggestion of emotion
    * 5-6: Some emotional content is present but unclear
    * 7-8: Clear emotional expression that is easily understood
    * 9-10: Powerful, unmistakable emotional communication
  - Creative originality:
    Rate the originality and creativity of the artistic approach.
    Scale: 1 (Generic/Unoriginal) - 10 (Highly Creative/Original)
  - Expressive use of mark-making:
    How effectively does the artist use different types of marks (thick/thin lines, smooth/rough textures, fast/slow strokes) to create expressive content?
    Scale: 1 (Not Effective) - 10 (Highly Effective)
- **Overall**
  - Overall artistic merit:
    Considering all aspects, how would you rate the overall artistic quality of this work?
    Scale: 1 (Very Poor) - 10 (Exceptional)

E.2 Subjective Feedback

This subsection reports the subjective measures collected directly from participants. The questions targeted self-assessed expressiveness, emotional communication, and creative confidence, complementing expert evaluations with first-person accounts of change.

- **Expressiveness**
  - How expressive do you feel your drawings were today?
  - How satisfied are you with the emotional quality of your drawings today?
  - How well do you think your drawings communicated the intended emotions/feelings?
- **Creative confidence**
  - How confident do you feel about creating expressive digital art after today's session
  - Compared to Session 1 (pre-intervention), how has your creative confidence changed?

E.3 Cognitive Load Questionnaire

In this subsection, we document the customized NASA-TLX workload assessment administered across different intervention groups. The questionnaire items varied slightly between groups to align with their respective task demands (feedback, prompts, or both), enabling us to capture perceived cognitive effort under distinct learning conditions (on a 5-point Likert scale).

- **G2: Ideation prompts only**
  - How mentally demanding was it to understand and follow the ideation prompts during the drawing tasks?
  - How much effort did you need to interpret what each prompt was asking you to express?

  - To what extent did the prompts make you rethink how to map your emotions or ideas into your strokes?
  - Did the prompts ever feel overwhelming, confusing, or distracting while you were drawing?
- **G3: Motor feedback only**
  - How mentally demanding was it to understand and follow the motor feedback during the drawing tasks?
  - How much effort did you need to interpret what each feedback was asking you to express?
  - How hard was it to adjust your stroke on the next attempt based on the feedback you received?
  - Did the feedback ever feel overwhelming, confusing, or distracting while you were drawing?
- **G4: Motor feedback and Ideation prompts**
  - How mentally demanding was it to understand and follow the motor feedback and ideation prompts during the drawing tasks?
  - How much effort did you need to interpret what each feedback and prompt was asking you to express?
  - How hard was it to adjust your stroke on the next attempt based on the feedback you received?
  - To what extent did the prompts make you rethink how to map your emotions or ideas into your strokes?
  - Did the feedback and prompts ever feel overwhelming, confusing, or distracting while you were drawing?